

Summary

Software Engineer with strong CS fundamentals and full-stack production experience. Re-architected backend systems to cut query latency by over 80%, and independently designed and shipped a production tool now serving 500+ daily active users.

IIIT Trichy | Bachelors of Technology (B.Tech.) | 2020-2024

Technical Skills

- **Languages** — C/C++, Java, TypeScript, JavaScript, Python
- **Frontend** — ReactJS, NextJS, ThreeJS, WebGL, Shaders, GSAP
- **Backend & APIs** — NodeJS, ExpressJS, Flask, REST APIs, JWT/Auth 2.0, WebSockets
- **Databases** — MongoDB, PostgreSQL, Procedural Language SQL, Firebase Console
- **Cloud Integrations & DevOps** — AWS (EC2/ Lambda), Cloudflare, Git, RazorPay
- **AI Tools** — OpenAI API, Anthropic Claude, Prompt Engineering, AI-assisted Development

Experience

Systems Engineer — TCS (Nov 2024 – Present)

- Redesigned legacy SOAP-based transaction logic into REST APIs, decoupling core services and reducing coupling across the enterprise middleware stack.
- Re-architected backend callback logic with asynchronous, multi-threaded processing, cutting database query latency from 5+ seconds to sub-second (>80% reduction).
- Independently designed and shipped a full-stack internal tool (PDN Tracker) end-to-end, including RBAC and deployment pipeline, replacing a deprecated vendor tool — now in production with 500+ daily active users.

Projects

PDN Tracker - Enterprise Deployment Management System | *ReactJS, TailwindCSS, NodeJS, PostgreSQL/SpringBoot*

- Designed a state machine to model multi-stage deployment lifecycles with rollback and audit logging.
- Rebuilt the legacy Java backend as a Node.js/Express/PostgreSQL, removing a hosting dependency and improving deploy velocity.
- Implemented session-based authentication (35% fewer login failures) and RBAC for secure cross-team access.

Chunky Discord — Zero-Cost Distributed File Storage | *ReactJS, NodeJS, Firebase, AWS EC2*

- Designed a chunking and hash-based reassembly algorithm guaranteeing 100% file integrity across a stateless, horizontally distributed storage layer.
- Built multi-part upload/download and retry logic against a third-party API used as an experimental transport backend.
- Deployed on AWS EC2 with Firebase auth as a system-design exercise in statelessness and data-integrity guarantees.

Competitive Programming

LeetCode rating 1834 (Knight) — 500+ problems solved across Data Structures and algorithmic challenges